

IGDTUW ANVESHAN FOUNDATION

6 WEEKS INTERNSHIP ON MACHINE LEARNING AND GENERATIVE AI USING PYTHON

College: Indira Gandhi Delhi Technical University for Women

Program: B. TECH

Department: Computer Science and Engineering

Enrollment Number: 23201012025

Student Name: Yumna Nasir

ASSIGNMENT 1

INPUT:

```
intern1.py - C:\Users\Yumna Nasir\OneDrive\Desktop\intern1.py (3.14.2)
File Edit Format Run Options Window Help

#ASSIGNMENT 1

#1. Find area of rectangle.
x = int(input("Enter length of rectangle:"))
y = int(input("Enter breadth of rectangle:"))
res = x * y
print("Area of rectangle is",res)

#2. Find simple interest.
def fun(p, t, r):
    return (p * t * r) / 100
p = int(input("Enter principal amount:"))
t = int(input("Enter time period:"))
r = int(input("Enter rate of interest:"))
res = fun(p, t, r)
print("Simple interest is", res)

#3. Convert temperature from Celsius to Fahrenheit.
c = float(input("Enter temperature in Celsius: "))
f = (c * 1.8) + 32
print("Temperature in Celsius", str(c))
print("Temperature in Fahrenheit", str(f))

#4. Calculate average of 3 numbers.
num1 = int(input("Enter value 1:"))
num2 = int(input("Enter value 2:"))
num3 = int(input("Enter value 3:"))
sum_numbers = num1 + num2 + num3
average = sum_numbers / 3
print(f'Sum of 3 numbers: {sum_numbers}, Average of 3 numbers: {average}')

#5. Find square and cube of a number.
n = int(input("Enter the value: "))
square = n ** 2
cube = n ** 3
print("Square of the number :", square)
print("Cube of the number :", cube)

#6. Swap two numbers without third variable.
x, y = 5, 7
print("Before swapping:", x, y)
x, y = y, x
print("After swapping:", x, y)
```

```
#7. Create a Student Report Program that take student details using input(), store marks in variables, calculate total and percentage , use comments, use proper indentation
name = input("Enter student's name: ")
roll_no = input("Enter student's roll number: ")
student_class = input("Enter student's class: ")
division = input("Enter student's division: ")
print("\nEnter marks for the following subjects (out of 100):")
maths_marks = float(input("Maths: "))
english_marks = float(input("English: "))
science_marks = float(input("Science: "))
social_science_marks = float(input("Social Science: "))
it_marks = float(input("IT: "))
total_marks = maths_marks + english_marks + science_marks + social_science_marks + it_marks
number_of_subjects = 5
average_marks = total_marks / number_of_subjects
percentage = total_marks / 500 * 100
print("\n--- Student Report ---")
print(f"Name: {name}")
print(f"Roll Number: {roll_no}")
print(f"Class: {student_class}")
print(f"Division: {division}")
print(f"Total Marks: {total_marks:.2f}")
print(f"Average Marks: {average_marks:.2f}")
print(f"Percentage: {percentage:.2f}")
```

OUTPUT:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Yumna Nasir\OneDrive\Desktop\internl.py =====
Enter length of rectangle:13
Enter breadth of rectangle:18
Area of rectangle is 234
Enter principal amount:196
Enter time period:4
Enter rate of interest:2
Simple interest is 15.68
Enter temperature in celsius: 42
Temperature in Celsius 42.0
Temperature in Fahrenheit 107.60000000000001
Enter value 1:13
Enter value 2:23
Enter value 3:24
Sum of 3 numbers: 60, Average of 3 numbers: 20.0
Enter the value: 91
Square of the number : 8281
Cube of the number : 753571
Before swapping: 5 7
After swapping: 7 5
Enter student's name: Khwarizmi
Enter student's roll number: 1
Enter student's class: 11
Enter student's division: A

Enter marks for the following subjects (out of 100):
Maths: 100
English: 100
Science: 100
Social Science: 100
IT: 100

--- Student Report ---
Name: Khwarizmi
Roll Number: 1
Class: 11
Division: A
Total Marks: 500.00
Average Marks: 100.00
Percentage: 100.00
>>>
```